Negative Binomial and Zero-Inflated Negative Binomial Regression by IRLS and EM

STA 651 Module 2 Final Project: the nbzinbPate package

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1 Introduction

Many outcomes in health, the social sciences and industry are counts: the number of doctor visits in a year, of articles a student publishes, of defects on a production line. The Poisson regression model is the natural starting point, but it forces the variance to equal the mean. Real count data are often **overdispersed**, with a variance much larger than the mean, and can also contain **more zeros** than a Poisson model with the same mean can produce. Ignoring overdispersion gives standard errors that are too small, and ignoring excess zeros gives a poor description of who has a zero count and why.

The negative binomial (NB2) model handles overdispersion by adding a dispersion parameter θ, so that $\mathrm{Var}(Y) = \mu + \mu^2/\theta$. The zero-inflated negative binomial (ZINB) model goes further and treats the data as coming from two processes: a *structural* process that produces zeros only (for example, people who never see a doctor), and an NB count process that can also produce zeros.

The goal of this project is an R package, **nbzinbPate**, that fits both models with estimation code written from scratch. The NB model is fitted by iteratively reweighted least squares (IRLS) for the regression coefficients, alternated with Newton-Raphson for $\log\theta$. The ZINB model, whose likelihood is not that of a generalized linear model, is fitted by an EM algorithm whose M-steps are IRLS fits. No general-purpose optimizer (`optim()`, `nlm()`, …) and no GLM fitter (`glm()`, `lm()`, …) is used. The package is checked against `MASS::glm.nb()` and `pscl::zeroinfl()`, studied in a simulation, and applied to physician office visits in the NMES1988 data.

2 Methods

2.1 Models

NB2. For counts Y_i with covariates x_i and offset o_i,

$$P(Y_i = y) = \frac{\Gamma(y+\theta)}{\Gamma(\theta)\,y!}\left(\frac{\theta}{\theta+\mu_i}\right)^{\theta}\left(\frac{\mu_i}{\theta+\mu_i}\right)^{y}, \qquad \log\mu_i = x_i^{\top}\beta + o_i,$$

so that $E(Y_i) = \mu_i$ and $\mathrm{Var}(Y_i) = \mu_i + \mu_i^2/\theta$. As $\theta \to \infty$ the model reduces to the Poisson.

ZINB. With probability π_i, where $\mathrm{logit}\,\pi_i = z_i^\top \gamma$, observation i is a structural zero; otherwise it comes from $\mathrm{NB}(\mu_i, \theta)$:

$$P(Y_i = 0) = \pi_i + (1-\pi_i) f_{NB}(0; \mu_i, \theta), \qquad P(Y_i = y) = (1-\pi_i) f_{NB}(y; \mu_i, \theta),\ y \geq 1,$$

where $f_{NB}(0; \mu, \theta) = \{\theta/(\theta+\mu)\}^\theta$. The observed-data log-likelihood is

$$\ell(\beta, \gamma, \theta) = \sum_{y_i = 0} \log[\pi_i + (1-\pi_i) f_{NB}(0; \mu_i, \theta)] + \sum_{y_i > 0} [\log(1-\pi_i) + \log f_{NB}(y_i; \mu_i, \theta)].$$

Because every zero contributes the log of a *sum*, ℓ is not the log-likelihood of an exponential-family GLM, and IRLS cannot be applied to it directly.

2.2 NB: IRLS for β alternated with Newton-Raphson for $\log\theta$

IRLS with θ fixed. For the log link, $\partial\mu/\partial\eta = \mu$, and $V(\mu) = \mu + \mu^2/\theta$. Fisher scoring is then a weighted least-squares regression of a working response z on X:

$$w_i = a_i \frac{(\partial\mu_i/\partial\eta_i)^2}{V(\mu_i)} = a_i \frac{\theta\mu_i}{\theta+\mu_i}, \qquad z_i = (\eta_i - o_i) + \frac{y_i - \mu_i}{\mu_i}, \qquad \beta^{(t+1)} = (X^\top W X)^{-1} X^\top W z,$$

where a_i are prior weights ($a_i = 1$ for an ordinary fit). Each update is solved by a QR decomposition of $W^{1/2}X$, which avoids forming $X^\top W X$ and squaring its condition number.

Newton-Raphson with β fixed. The score and second derivative of ℓ in θ are

$$U(\theta) = \sum_i a_i \Big[\psi(y_i+\theta) - \psi(\theta) + \log\theta + 1 - \log(\theta+\mu_i) - \frac{y_i+\theta}{\theta+\mu_i}\Big],$$

$$H(\theta) = \sum_i a_i \Big[\psi'(y_i+\theta) - \psi'(\theta) + \frac{1}{\theta} - \frac{2}{\theta+\mu_i} + \frac{y_i+\theta}{(\theta+\mu_i)^2}\Big],$$

with ψ and ψ' the digamma and trigamma functions. The update is done on $\phi = \log\theta$, which keeps $\theta > 0$; by the chain rule

$$U_\phi = \theta U, \qquad H_\phi = \theta^2 H + \theta U, \qquad \phi \leftarrow \phi - U_\phi / H_\phi.$$

Outer loop. The fit starts from a Poisson IRLS fit for β (the same IRLS code with $\theta = \infty$, so $w_i = a_i\mu_i$) and the moment estimate $\theta_0 = \sum_i a_i / \sum_i a_i (y_i/\mu_i - 1)^2$, then alternates the two steps, warm-starting each one at the current values.

Standard errors. At convergence, $\widehat{\mathrm{Var}}(\hat\beta) = (X^\top \hat W X)^{-1}$, treating θ as known as `glm.nb()` does, and $\mathrm{SE}(\hat\theta) = \{-H(\hat\theta)\}^{-1/2}$.

2.3 ZINB: EM with IRLS M-steps

Let $D_i = 1$ if observation i is a structural zero. If the D_i were observed, the complete-data log-likelihood would split into a logistic regression for D_i and a weighted NB regression with weights $1 - D_i$, each of which IRLS can fit. EM replaces D_i by its conditional expectation.

E-step. The posterior probability of a structural zero is

$$\tau_i = \frac{\pi_i}{\pi_i + (1-\pi_i) f_{NB}(0; \mu_i, \theta)} \quad (y_i = 0), \qquad \tau_i = 0 \quad (y_i > 0).$$

M-step, zero part. Maximize $\sum_i [\tau_i \log \pi_i + (1-\tau_i) \log(1-\pi_i)]$ over γ: a logistic regression with fractional response τ_i, fitted by IRLS with $w_i = \pi_i(1-\pi_i)$ and $z_i = z_i^\top \gamma + (\tau_i - \pi_i)/\{\pi_i(1-\pi_i)\}$. Because the logit is the canonical link, this IRLS is exact Newton-Raphson.

M-step, count part. Maximize $\sum_i (1-\tau_i) \log f_{NB}(y_i; \mu_i, \theta)$ over (β, θ). This is exactly the NB fit above with prior weights $a_i = 1 - \tau_i$, so the same code is reused.

Both M-steps are run to convergence, warm-started at the previous values. EM starts from the NB fit for (β, θ) and a logistic IRLS fit of $I(y_i = 0)$ for γ. Each M-step increases the expected complete-data log-likelihood, which guarantees that ℓ never decreases (Dempster, Laird and Rubin, 1977); the code checks this at every iteration.

2.4 Standard errors for ZINB: analytic Hessian

EM does not produce standard errors, so after convergence the observed information of ℓ with respect to $(\beta, \gamma, \log\theta)$ is computed analytically and inverted. Writing $\eta_i^z = z_i^\top \gamma$, $\eta_i^c = x_i^\top \beta + o_i$, $\ell_i^f = \log f_{NB}(y_i; \mu_i, \theta)$ and $a, b \in \{\eta^c, \phi\}$, the derivatives simplify when expressed through τ_i:

$$\frac{\partial \ell_i}{\partial \eta_i^z} = \tau_i - \pi_i, \qquad \frac{\partial \ell_i}{\partial a} = (1-\tau_i) \frac{\partial \ell_i^f}{\partial a},$$

$$\frac{\partial^2 \ell_i}{\partial (\eta_i^z)^2} = \tau_i(1-\tau_i) - \pi_i(1-\pi_i), \quad \frac{\partial^2 \ell_i}{\partial \eta_i^z \, \partial a} = -\tau_i(1-\tau_i) \frac{\partial \ell_i^f}{\partial a},$$

$$\frac{\partial^2 \ell_i}{\partial a \, \partial b} = (1-\tau_i) \frac{\partial^2 \ell_i^f}{\partial a \, \partial b} + \tau_i(1-\tau_i) \frac{\partial \ell_i^f}{\partial a} \frac{\partial \ell_i^f}{\partial b}.$$

These are Louis' (1982) formula: the observed information equals the expected complete-data information minus the variance of the complete-data score, and $\tau_i(1-\tau_i) = \mathrm{Var}(D_i \mid y_i)$ is the "missing information". The NB derivatives $\partial \ell_i^f / \partial \eta_i^c = \theta(y_i - \mu_i)/(\theta + \mu_i)$, $\partial^2 \ell_i^f / \partial (\eta_i^c)^2 = -\theta\mu_i(\theta + y_i)/(\theta + \mu_i)^2$, $\partial^2 \ell_i^f / \partial \eta_i^c \partial \phi = \theta\mu_i(y_i - \mu_i)/(\theta + \mu_i)^2$, and the ϕ derivatives θU_i and $\theta U_i + \theta^2 H_i$ come from Section 2.2. The chain rule ($\partial \eta_i^c / \partial \beta = x_i$, $\partial \eta_i^z / \partial \gamma = z_i$) assembles the full Hessian H, and $\widehat{\mathrm{Var}}(\hat\beta, \hat\gamma, \log\hat\theta) = \{-H\}^{-1}$. The standard error of $\hat\theta$ follows from the delta method, $\mathrm{SE}(\hat\theta) = \hat\theta \, \mathrm{SE}(\log\hat\theta)$.

2.5 Convergence criteria and numerical safeguards

The inner IRLS loops stop when no coefficient changes by more than 10^{-10}, and the Newton loop when $|\Delta\phi| < 10^{-10}$. A criterion on the parameters is needed because the log-likelihood is flat at its maximum: a change in ℓ of 10^{-7} can leave β wrong in the sixth digit, and the NB log link is not canonical, so Fisher scoring converges only linearly. The NB outer loop stops when ℓ changes by less than 10^{-8} *and* the parameters by less than 10^{-8}; EM stops when ℓ changes by less than 10^{-8} and the parameters by less than 10^{-6}.

Each IRLS or Newton step is halved if it lowers the log-likelihood by more than rounding error which means it passed the maximum. The steps are halved until the maximum is not passed. Probabilities are handled on the log scale (`plogis(eta, log.p = TRUE)` and a log-sum-exp form of $\log[\pi + (1-\pi)f_{NB}(0)]$), so π_i is never rounded to exactly 0 or 1. Two boundary cases are detected and reported with a warning: $\hat{\theta}$ diverging to ∞ (no overdispersion; θ is capped at 10^8), and all $\hat{\pi}_i \to 0$ in the ZINB model (no zero inflation, so the zero-model coefficients diverge to $-\infty$).

3 The R package

The package exports two fitting functions, two control functions, and a full set of S3 methods for each fitted class.

Function	Purpose
`nbreg(formula, data, offset, control)`	NB2 regression: IRLS + Newton-Raphson for $\log\theta$
`zinbreg(formula, data, offset, control)`	ZINB regression by EM; formula `y ~ x | z`
`nbreg_control()`, `zinbreg_control()`	tolerances, iteration limits, `trace`
S3 methods (both classes)	`print`, `summary`, `coef`, `vcov`, `logLik`, `AIC`/`BIC`, `fitted`, `residuals`, `nobs`, `predict`

Design. Each model has a formula interface (`nbreg()`, `zinbreg()`) that builds the design matrices, checks the input and packages the result, and a matrix-level engine (`nbreg_fit()`, `zinbreg_fit()`) that only does the estimation. This separation is what makes the EM count-part M-step a single call to `nbreg_fit()` with prior weights $1 - \tau_i$ and a warm start. The two-part formula `y ~ x | z` is split in base R: a formula is a call, and because `|` binds more loosely than `+`, its right-hand side is a call to `|` whose two arguments are the count and zero parts. Without `|`, the same covariates are used in both parts, as in `zeroinfl()`. Offsets may be given as an argument or as `offset()` terms in the (count) formula. The only dependency is base R's **stats** package.

Methods. `logLik()` returns an object carrying the number of parameters and observations, so `AIC()` and `BIC()` work. `residuals()` returns response or Pearson residuals; for ZINB the Pearson residuals use $\mathrm{Var}(Y_i) = (1-\pi_i)\mu_i\{1 + \mu_i(\pi_i + 1/\theta)\}$. `predict()` accepts `newdata` (using the fitted factor levels, contrasts and offset) and, for ZINB, `type = "response"` ($(1-\pi)\mu$), `"count"` (μ) or `"zero"` (π).

Testing. The `testthat` suite compares both functions with `glm.nb()` and `zeroinfl()` at the project tolerances on both data sets, checks the analytic score and Hessian against numerical derivatives, checks that EM never decreases ℓ, tests offsets, single-covariate and intercept-only

models, the boundary cases and informative errors, and verifies that no function in the package calls a prohibited optimizer or fitter. `R CMD check` passes with no errors, warnings or notes on Windows, macOS and Linux through GitHub Actions. The package is installed with `devtools::install_github("brandonp8/nbzinbPate")`; Section 6.2 shows it in use, and the README has a further example.

4 Benchmarking

Both functions were compared with the reference implementations on `pscl::bioChemists` and `AER::NMES1988`, using the reference functions with tight tolerances (`glm.control(epsilon = 1e-10)` and `zeroinfl.control(reltol = 1e-12)`). The script is `inst/benchmark/benchmark.R`.

Table 1: Maximum absolute difference from MASS::glm.nb() (NB) and pscl::zeroinfl() (ZINB), median time, and number of outer (NB) or EM (ZINB) iterations. Required: at most 1e-5 (NB coefficients and SEs), 1e-4 (NB theta, logLik), 1e-3 (all ZINB).

Data	Model	Coef.	SE	theta / log theta	logLik	Ours (s)	Ref. (s)	Iter.
bioChemists	NB	1.9e-11	9.4e-13	2.6e-12	2.3e-13	0.03	0.05	5
bioChemists	ZINB	8.9e-06	5.4e-06	7.2e-07	1.6e-10	2.36	0.16	140
NMES1988	NB	7.2e-11	4.3e-12	1.4e-10	0.0e+00	0.12	0.14	4
NMES1988	ZINB	5.6e-06	2.8e-06	1.0e-06	8.9e-10	4.17	0.80	64

All quantities meet the required tolerances, by a wide margin. For NB, the coefficients, standard errors, θ and log-likelihood agree with `glm.nb()` to within 1.4e-10, essentially machine precision. (The only slightly larger difference is in $\mathrm{SE}(\widehat{\theta})$, about 1.1e-06: `MASS::theta.ml()` evaluates the information at the θ value before its final Newton step, while `nbreg()` evaluates it at $\widehat{\theta}$.) For ZINB the largest difference in any coefficient or standard error is 8.9e-06, and the log-likelihoods agree to 8.9e-10. On NMES1988 our ZINB log-likelihood is slightly higher than that of `zeroinfl()`. The complete side-by-side estimates are in the Appendix.

The NB fits take about as long as `glm.nb()`. The ZINB fits are 5 to 15 times slower than `zeroinfl()`. This is expected. `zeroinfl()` updates all parameters jointly with a quasi-Newton method (BFGS), which converges superlinearly, whereas EM converges linearly at a rate set by the fraction of missing information. The zero component is weakly identified in bioChemists, so EM needs 140 iterations there, against 64 for the larger NMES1988 data. In addition, every M-step is a fully converged weighted NB fit.

5 Simulation study

To check that `zinbreg()` recovers known parameters and that its standard errors are accurate, $R = 100$ data sets of size $n = 500$ were simulated (seed 651, script `inst/benchmark/simulation.R`) from $\log \mu_i = 1 + 0.5\, x_{1i} - 0.3\, x_{2i}$ with $x_1 \sim N(0, 1)$ and $x_2 \sim \text{Bernoulli}(0.5)$, $\text{logit}\, \pi_i = -0.5 + 0.8\, z_{1i}$

with $z_1 \sim N(0,1)$, and $\theta = 1.5$. Covariates were redrawn for each data set. On average about 55% of each simulated sample were zeros.

Table 2: Simulation results: bias, Monte Carlo SD versus mean estimated SE, and empirical coverage of 95% Wald intervals (theta on the log scale).

Parameter	True	Mean est.	Bias	MC SD	Mean SE	SE / MC SD	Coverage
count_(Intercept)	1.000	0.997	-0.003	0.098	0.107	1.094	0.97
count_x1	0.500	0.509	0.009	0.060	0.067	1.111	0.95
count_x2	-0.300	-0.311	-0.011	0.114	0.131	1.145	0.97
zero_(Intercept)	-0.500	-0.526	-0.026	0.273	0.240	0.879	0.94
zero_z1	0.800	0.833	0.033	0.222	0.191	0.862	0.96
Log(theta)	0.405	0.425	0.020	0.242	0.256	1.057	0.96

All 100 fits converged, with 0 boundary cases and no warnings. The estimates are essentially unbiased: every bias is within 1.50 Monte Carlo standard errors of zero (the Monte Carlo SE of a mean estimate is $\mathrm{SD}/\sqrt{R}$). The ratio of the mean estimated SE to the Monte Carlo SD ranges from 0.86 to 1.14. With $R = 100$, an SD is itself estimated with a relative error of about $1/\sqrt{2(R-1)} \approx 7\%$, so these ratios are consistent with 1. The empirical coverage of the 95% Wald intervals ranges from 0.94 to 0.97, all within two Monte Carlo standard errors ($\pm 2 \times 0.022$) of 0.95. This also provides an independent check of the analytic Hessian: errors in the second derivatives would show up as SE ratios far from 1. The standardized errors in the figure below are centred at 0 with roughly unit spread.

The only visible pattern is that the zero-part SEs are somewhat smaller than the Monte Carlo SDs (ratios 0.86 and 0.88), while the count-part SEs are slightly larger. With $R = 100$ these deviations are not statistically significant, but a slight underestimation of uncertainty in the weakly identified zero component at $n = 500$ is plausible.

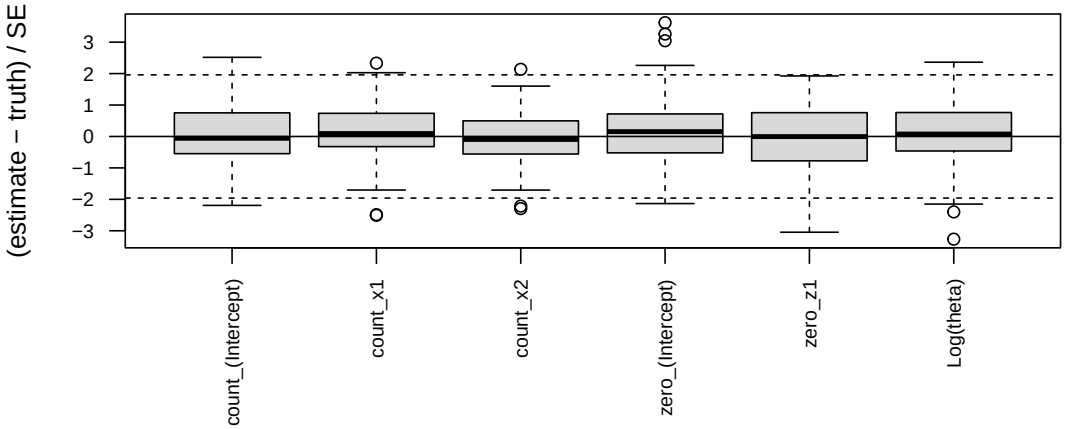


Figure 1: Standardized errors (estimate - truth) / SE across the 100 replications. Under correct SEs these are approximately N(0, 1); dashed lines at +/-1.96.

6 Data analysis: NMES1988

The NMES1988 data (Deb and Trivedi, 1997; available in the **AER** package) record the number of physician office visits (`visits`) for 4,406 individuals aged 66 and over in the US National Medical Expenditure Survey, 1987-88. The covariates used are the number of hospital stays (`hospital`), self-perceived health (`health`: poor, average or excellent), the number of chronic conditions (`chronic`), `gender`, years of education (`school`), and whether the individual has private insurance (`insurance`).

6.1 Descriptive summary

Table 3: Physician office visits in NMES1988. Poisson P(0): probability of a zero under a Poisson distribution with the sample mean.

n	Mean	Variance	Var./mean	Zeros	Prop. zero	Poisson P(0)
4406	5.774	45.687	7.912	683	0.155	0.003

The number of visits ranges from 0 to 89, with mean 5.77 and variance 45.69: the variance is 7.9 times the mean, far from the equality the Poisson model requires. There are also many more zeros than a Poisson distribution allows: 683 individuals (15.5%) made no office visit, while a Poisson distribution with the same mean gives a zero with probability 0.003, i.e. about 14 zeros. The covariates explain part of this, so the fitted models below give a fairer comparison, but the data clearly show overdispersion and excess zeros.

6.2 Models

```
f_count <- visits ~ hospital + health + chronic + gender + school + insurance
f_zinb  <- visits ~ hospital + health + chronic + gender + school + insurance |
  hospital + chronic + gender + school + insurance

m_pois <- glm(f_count, family = poisson, data = d)   # reference Poisson model
m_nb   <- nbreg(f_count, data = d)                     # our NB
m_zinb <- zinbreg(f_zinb, data = d)                    # our ZINB
```

The count part uses all six covariates. The zero part uses the minimum set required by the project (all covariates except self-perceived health). The Poisson model is fitted with `glm()` for reference only.

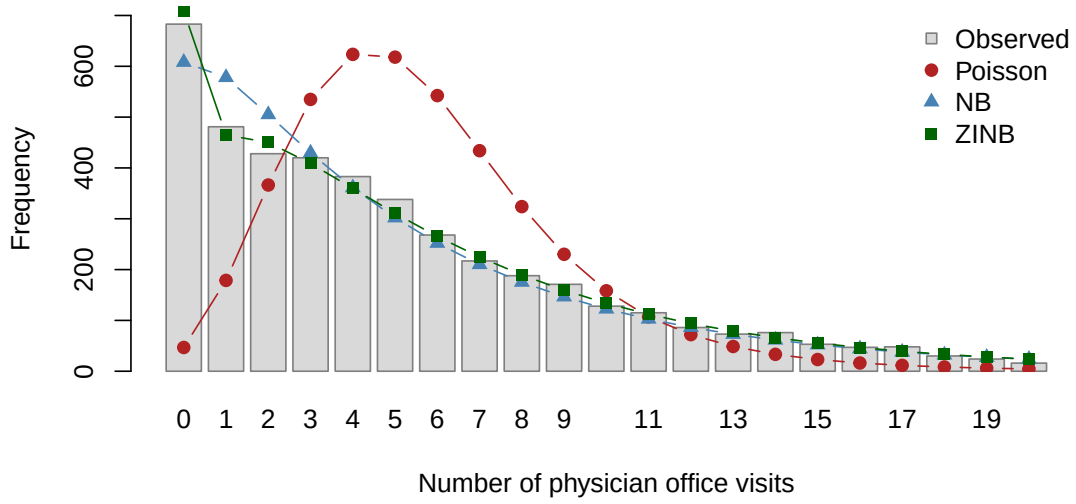


Figure 2: Observed frequencies of office visits (bars) and frequencies expected under the fitted Poisson, NB and ZINB models, for 0-20 visits.

6.3 Model comparison

Table 4: Model comparison for NMES1988 (observed zeros = 683).

Model	df	logLik	AIC	BIC	Expected zeros
Poisson	8	-17971.6	35959.2	36010.4	46.7
NB	9	-12170.6	24359.1	24416.6	608.0
ZINB	15	-12090.7	24211.4	24307.3	708.9

The Poisson model fits very poorly. It predicts only 47 zeros against 683 observed, and its log-likelihood is 5801 units below that of the NB model, for a single extra parameter. In the figure, it puts far too much mass at 3-8 visits and too little at 0-2 and in the right tail. Allowing overdispersion fixes most of this: the NB model has $\hat{\theta} = 1.207$ (SE 0.034), and its predicted frequencies follow the observed ones closely from about 3 visits upward.

The NB model still **under-predicts** the zeros (608 expected against 683 observed) and **over-predicts** single visits (578 against 481). The ZINB model corrects both (709 zeros and 465 single visits); it slightly over-predicts the zeros, by 26. It improves the log-likelihood by 79.8 with 6 extra parameters, and has both the lowest AIC (24211.4 against 24359.1 for NB) and the lowest BIC (24307.3 against 24416.6). Unlike the bioChemists example, where BIC preferred the simpler NB model, here both criteria favour zero inflation. Once the structural zeros are modelled separately, the count component is also less overdispersed ($\hat{\theta} = 1.483$ in the ZINB model).

6.4 Interpretation: incidence rate ratios and odds ratios

Table 5: Count part: incidence rate ratios with 95% Wald intervals. The reference health category is average (the contrasts supplied with the AER data).

Covariate	NB IRR (95% CI)	ZINB count IRR (95% CI)
hospital	1.24 (1.20, 1.29)	1.22 (1.18, 1.27)
healthpoor	1.36 (1.23, 1.49)	1.33 (1.22, 1.45)
healthexcellent	0.71 (0.63, 0.80)	0.73 (0.65, 0.82)
chronic	1.19 (1.16, 1.22)	1.14 (1.11, 1.16)
gendermale	0.88 (0.83, 0.94)	0.92 (0.87, 0.98)
school	1.03 (1.02, 1.04)	1.02 (1.01, 1.03)
insuranceyes	1.25 (1.16, 1.35)	1.13 (1.05, 1.23)

Table 6: ZINB zero part: odds ratios of being a structural zero (never visiting), with 95% Wald intervals.

Covariate	OR (95% CI)	p-value
hospital	0.45 (0.20, 1.02)	0.0571
chronic	0.29 (0.20, 0.41)	<1e-04
gendermale	1.91 (1.29, 2.83)	0.0012
school	0.92 (0.87, 0.97)	0.0014
insuranceyes	0.31 (0.20, 0.48)	<1e-04

All effects below hold the other covariates fixed.

NB model. Every covariate is associated with the number of visits at the 5% level. Each additional hospital stay is associated with 24% more office visits (IRR 1.24), and each additional chronic condition with 19% more. Compared with individuals in average health, those in poor health make 36% more visits and those in excellent health 29% fewer. Men make 12% fewer visits than women, each year of education is associated with 3% more visits, and private insurance with 25% more.

ZINB model, count part. This part describes individuals who are "at risk" of visiting a doctor. The directions and significance of all effects match the NB model, and the health and hospital effects are almost unchanged (IRR 1.22 per hospital stay; 1.33 and 0.73 for poor and excellent health). The effects of chronic conditions (IRR 1.19 to 1.14), insurance (1.25 to 1.13) and gender (0.88 to 0.92) move toward 1, although all remain significant. In the NB model these covariates combine two effects: whether someone sees a doctor at all, and how often. The ZINB model separates the two, and part of their association turns out to act through the first.

ZINB model, zero part. The zero part models the probability of being a structural zero, an individual who would not visit a physician regardless. Each additional chronic condition multiplies the odds of being such a non-user by 0.29, a 71% reduction, and private insurance multiplies them by 0.31 (69% lower odds). Men have 1.91 times the odds of women of being non-users, and each additional year of education lowers the odds by 8%. Hospital stays point in the same direction (OR 0.45), but this effect is not significant at the 5% level ($p = 0.057$). These effects are plausible: people with chronic conditions have a reason to see a doctor, and people with private insurance face lower costs of doing so.

Preferred model. I prefer the ZINB model for these data. It has the lowest AIC *and* BIC, it reproduces the observed numbers of zeros and single visits much better than the NB model, and its two parts have a clear and useful interpretation. Chronic illness, insurance, gender and education are associated both with *whether* an elderly person sees a physician at all and with *how often* among those who do, while the effect of hospital stays acts mainly on how often. (Self-perceived health was included in the count part only, so its role in non-use was not assessed.) The NB model is not wrong (all its effects have the same signs), but it mixes these two processes, and the comparison shows that the data support separating them.

7 Discussion

The package reproduces `glm.nb()` and `zeroinfl()` to far better than the required tolerances, using only IRLS, Newton-Raphson and EM. The simulation confirms that the ZINB estimates are unbiased and that the analytic-Hessian standard errors give close to nominal coverage. Several limitations remain.

- **Speed.** EM converges linearly, so `zinbreg()` is 5-15 times slower than `zeroinfl()`. Running each M-step to full convergence is also conservative. Loosening the M-step tolerance roughly halves the run time without changing the estimates, and EM acceleration (for example SQUAREM) could reduce the number of iterations further, although it needs extra safeguards to keep the log-likelihood monotone.
- **NB standard errors.** Following `glm.nb()`, $\mathrm{Var}(\hat{\beta})$ treats θ as known. Because β and θ are asymptotically orthogonal in the NB2 model, this costs little, but the full observed information would be more exact in small samples.
- **Weak identification.** When the zero component is weakly identified (large zero-part SEs, as in bioChemists), EM needs many iterations and the Wald intervals for γ may be somewhat too narrow, as the simulation hints. Profile-likelihood intervals would be more reliable there.
- **Boundary cases.** When the data show no zero inflation, the MLE of γ is at $-\infty$. The package detects this and warns, but it does not provide a formal test of zero inflation.
- **Extensions.** Natural extensions are zero-inflated Poisson and hurdle models, other links for the zero part (for example the probit), prior weights in `zinbreg()`, offsets in the zero model, and a test comparing the NB and ZINB models.

8 AI-use statement

Claude Code was used to generate all code and to draft the report. I gave it clear step by step instructions on how to implement the package as well as rules it must follow like not using optim() or other functions. After each step and then again after completion I double-checked its work. I had to make a few corrections to its work. For instance, at first the stopping rule for the IRLS was based on log-likelihood change which converged before beta converged. This is because the likelihood is nearly flat at its peak and the NB IRLS algorithm converges only linearly. I corrected this by having the algorithm stop when beta itself stopped changing.

9 References

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10 Appendix: side-by-side benchmark estimates

Table 7: bioChemists, NB: nbzinbPate vs MASS::glm.nb().

Quantity	Parameter	nbzinbPate	Reference	Abs. difference
Coefficients	(Intercept)	0.256144	0.256144	1.9e-11
Coefficients	femWomen	-0.216418	-0.216418	4.1e-12
Coefficients	marMarried	0.150489	0.150489	2.4e-12
Coefficients	kid5	-0.176415	-0.176415	3.9e-12
Coefficients	phd	0.015271	0.015271	7.2e-13
Coefficients	ment	0.029082	0.029082	1.6e-12
Standard errors	(Intercept)	0.137348	0.137348	9.4e-13
Standard errors	femWomen	0.072636	0.072636	4.0e-13
Standard errors	marMarried	0.082097	0.082097	5.1e-13
Standard errors	kid5	0.052813	0.052813	2.3e-13
Standard errors	phd	0.035873	0.035873	2.1e-13
Standard errors	ment	0.003214	0.003214	1.7e-14
theta	theta	2.264388	2.264388	2.6e-12
SE(theta)	SE(theta)	0.271176	0.271175	1.1e-06
Log-likelihood	logLik	-1560.958338	-1560.958338	2.3e-13

Table 8: bioChemists, ZINB: nbzinbPate vs pscl::zeroinfl().

Quantity	Parameter	nbzinbPate	Reference	Abs. difference
Coefficients	count_(Intercept)	0.416747	0.416747	1.8e-07
Coefficients	count_femWomen	-0.195507	-0.195507	1.5e-07
Coefficients	count_marMarried	0.097583	0.097583	6.3e-08
Coefficients	count_kid5	-0.151732	-0.151732	1.1e-07
Coefficients	count_phd	-0.000700	-0.000700	2.6e-08
Coefficients	count_ment	0.024786	0.024786	1.9e-08
Coefficients	zero_(Intercept)	-0.191696	-0.191688	7.3e-06
Coefficients	zero_femWomen	0.635933	0.635933	2.8e-07
Coefficients	zero_marMarried	-1.499460	-1.499468	8.9e-06
Coefficients	zero_kid5	0.628426	0.628427	1.1e-06
Coefficients	zero_phd	-0.037713	-0.037715	1.5e-06
Coefficients	zero_ment	-0.882289	-0.882293	4.6e-06
Standard errors	count_(Intercept)	0.143596	0.143597	3.3e-07
Standard errors	count_femWomen	0.075593	0.075593	8.9e-09
Standard errors	count_marMarried	0.084452	0.084452	4.6e-09
Standard errors	count_kid5	0.054206	0.054206	8.3e-09
Standard errors	count_phd	0.036270	0.036270	8.1e-08
Standard errors	count_ment	0.003492	0.003493	2.5e-07
Standard errors	zero_(Intercept)	1.322815	1.322819	3.8e-06
Standard errors	zero_femWomen	0.848914	0.848918	4.0e-06
Standard errors	zero_marMarried	0.938665	0.938671	5.4e-06
Standard errors	zero_kid5	0.442781	0.442783	1.5e-06
Standard errors	zero_phd	0.308007	0.308008	8.8e-07
Standard errors	zero_ment	0.316226	0.316228	2.6e-06
log(theta)	log(theta)	0.976357	0.976357	7.2e-07
SE(log theta)	SE(log theta)	0.135468	0.135469	1.5e-06
Log-likelihood	logLik	-1549.990887	-1549.990887	1.6e-10

Table 9: NMES1988, NB: nbzinbPate vs MASS::glm.nb().

Quantity	Parameter	nbzinbPate	Reference	Abs. difference
Coefficients	(Intercept)	0.929257	0.929257	5.8e-11
Coefficients	hospital	0.217772	0.217772	7.2e-11
Coefficients	healthpoor	0.305013	0.305013	3.8e-11
Coefficients	healthexcellent	-0.341807	-0.341807	2.1e-11
Coefficients	chronic	0.174916	0.174916	1.7e-11
Coefficients	gendermale	-0.126488	-0.126488	9.2e-12
Coefficients	school	0.026815	0.026815	7.2e-13
Coefficients	insuranceyes	0.224402	0.224402	2.1e-12
Standard errors	(Intercept)	0.054591	0.054591	3.9e-12
Standard errors	hospital	0.020176	0.020176	5.8e-13
Standard errors	healthpoor	0.048511	0.048511	2.4e-12
Standard errors	healthexcellent	0.060924	0.060924	4.3e-12

Quantity	Parameter	nbzinbPate	Reference	Abs. difference
Standard errors	chronic	0.012092	0.012092	7.4e-13
Standard errors	gendermale	0.031216	0.031216	2.2e-12
Standard errors	school	0.004394	0.004394	3.1e-13
Standard errors	insuranceyes	0.039464	0.039464	2.7e-12
theta	theta	1.206604	1.206604	1.4e-10
SE(theta)	SE(theta)	0.033555	0.033555	4.2e-07
Log-likelihood	logLik	-12170.553598	-12170.553598	0.0e+00

Table 10: NMES1988, ZINB: nbzinbPate vs pscl::zeroinfl().

Quantity	Parameter	nbzinbPate	Reference	Abs. difference
Coefficients	count_(Intercept)	1.193713	1.193713	5.7e-07
Coefficients	count_hospital	0.201477	0.201477	2.6e-08
Coefficients	count_healthpoor	0.285133	0.285133	8.9e-08
Coefficients	count_healthexcellent	-0.319336	-0.319336	2.0e-07
Coefficients	count_chronic	0.129000	0.129000	1.5e-07
Coefficients	count_gendermale	-0.080277	-0.080277	1.5e-07
Coefficients	count_school	0.021423	0.021423	2.4e-08
Coefficients	count_insuranceyes	0.125863	0.125864	2.5e-07
Coefficients	zero_(Intercept)	-0.046884	-0.046879	5.6e-06
Coefficients	zero_hospital	-0.800487	-0.800483	3.3e-06
Coefficients	zero_chronic	-1.247933	-1.247938	5.4e-06
Coefficients	zero_gendermale	0.647678	0.647678	4.6e-07
Coefficients	zero_school	-0.083770	-0.083771	6.6e-07
Coefficients	zero_insuranceyes	-1.175611	-1.175615	4.2e-06
Standard errors	count_(Intercept)	0.056661	0.056661	1.6e-07
Standard errors	count_hospital	0.020360	0.020360	8.3e-09
Standard errors	count_healthpoor	0.045093	0.045093	2.9e-09
Standard errors	count_healthexcellent	0.060405	0.060405	6.4e-09
Standard errors	count_chronic	0.011931	0.011931	5.7e-09
Standard errors	count_gendermale	0.031024	0.031024	8.5e-09
Standard errors	count_school	0.004358	0.004358	2.0e-08
Standard errors	count_insuranceyes	0.041588	0.041588	9.6e-09
Standard errors	zero_(Intercept)	0.268561	0.268558	2.8e-06
Standard errors	zero_hospital	0.420817	0.420816	7.2e-07
Standard errors	zero_chronic	0.178316	0.178316	5.9e-07
Standard errors	zero_gendermale	0.200114	0.200115	7.8e-07
Standard errors	zero_school	0.026254	0.026253	7.1e-07
Standard errors	zero_insuranceyes	0.220119	0.220121	2.1e-06
log(theta)	log(theta)	0.394144	0.394143	1.0e-06
SE(log theta)	SE(log theta)	0.035035	0.035035	9.3e-08
Log-likelihood	logLik	-12090.722012	-12090.722012	8.9e-10